Expenses Reporting Tool for an IT Consultancy

Project Management (GEP)
Assignment 3: Budget and Sustainability Analysis

Bachelor Degree in Informatics Engineering
Specialization in Software Engineering

Bachelor's Thesis

Natalia Yike Wang Yang

Thesis Supervisor
Sergio Alejandro Lobig Machado (FSP Consulting Services Limited)

Thesis Director
Marc Oriol Hilari (Department of Software Engineering)

GEP Tutor
Eduard Balbuena Longo (Department of Project Management)

Universitat Politècnica de Catalunya
Facultat Informàtica de Barcelona
Computer Science Department

March 8, 2026

Contents

List of Figures

List of Tables

6 Budget and Sustainability

The final fundamental aspect for concluding the initial documentation of a project is the budget and the project's economic viability. In addition, it is important to analyse the project's sustainability, considering the impact of the proposed solution on economic, environmental, and social dimensions. This stage studies both positive and negative project consequences.

6.1 Budget

This section details all project expenses, justifying each cost and its corresponding calculation. The ultimate goal is to determine the initial project budget.

6.1.1 Personnel Cost

To obtain realistic personnel cost estimates, the average hourly wage for each personnel member was calculated in advance using market salary data[**annual-spain-salary**]. This calculation assumes 260 working days per year with an average of 7.5 working hours per day.

In Table 6.1.1, the hourly wage calculation for each member of personnel is specified. Note that the final number already includes the Social Security (SS) cost. Also note that personnel are categorized by roles rather than by the human resources (HR) identified in Section 5.2.1. Each resource is assigned to at least one role in the table. There are five different roles involved in this project:

- **Project Manager (PM)**: Responsible for planning and leading the entire project.
- **Junior Full-Stack Developer (SE)**: Responsible for the full coding implementation of the application
- **QA Tester (T)**: Responsible for verifying the applications' functionality.
- **Software Architect (SA)**: Responsible for designing the application's technical structure while considering the project requirements and scope.
- **Technical Writer (TW)**: Responsible for elaborating a clear and accurate project documentation.

Role	**Annual Salary (€)**	**Total including SS (€)**	**Cost / Hour (€)**	**Assigned HR**
Project Manager	43,300	56,290	28.87	SM, PO, TD, GT
Full-stack Junior Developer	29,800	38,740	19.87	DT
QA Tester	26,888	34,954.4	17.93	DT
Software Architect	41,800	54,340	27.87	DT
Technical Writer	31,300	40,690	20.87	DT

Table 6.1.1: Annual and hourly salaries for different project roles. [Own creation]

In Table 6.1.2, personnel costs for each task and role, as well as the overall cost, are calculated. These costs result from multiplying each personnel member's effort hours (from Table 5.3.1) by their corresponding role's hourly wage (from Table 6.1.1).

			Hours per role					
ID	**Name**	**Total hours**	PM	D	T	SA	TW	**Cost (€)**
PS	**Project Starting**	**50**	**80**	**0**	**0**	**0**	**0**	**2,309.33**
PS1	Context and Scope	20	20	0	0	0	0	577.33
PS2	Temporal Planning	10	20	0	0	0	0	577.33
PS3	Budget and Sustainability	10	20	0	0	0	0	577.33
PS4	Initial Documentation	10	20	0	0	0	0	577.33
PM	**Project Management**	**153**	**59**	**112**	**0**	**0**	**41**	**4,783.73**
PM1	Agile Ceremonies	51	51	51	0	0	0	2,485.40
PM2	Thesis Director Review Meetings	8	8	8	0	0	0	389.87
PM3	Git Repository Documentation	40	0	40	0	0	0	794.67
PM4	User Manual Documentation	2	0	1	0	0	1	40.73
PM5	Product Backlog Update	8	0	8	0	0	0	158.93
PM6	Final Thesis Documentation	20	0	0	0	0	20	417.33
PM7	Final Demonstration Video	4	0	4	0	0	0	79.47
PM8	Bachelor's Thesis Defense	20	0	0	0	0	20	417.33
DT	**Design**	**16**	**0**	**0**	**0**	**16**	**0**	**445.87**
DT1	Use Case Diagram	2	0	0	0	2	0	55.73
DT2	Conceptual Model Diagram	4	0	0	0	4	0	111.47
DT3	Technology Stack Decision	2	0	0	0	2	0	55.73
DT4	Logic and Physic Architecture Design	4	0	0	0	4	0	111.47
DT5	UI mockups Design	2	0	0	0	2	0	55.73
DT6	Security Design	2	0	0	0	2	0	55.73
Dev	**Development**	**36**	**0**	**31**	**5**	**0**	**0**	**705.49**
Dev0	Self-learning and study of unfamiliar technologies	8	0	8	0	0	0	158.93
Dev1	Environment Configuration	2	0	2	0	0	0	39.73
Dev2	Product Deployment	8	0	8	0	0	0	158.93
Dev3	Unit and Integration Testing	10	0	5	5	0	0	188.96
Dev4	Logging and Monitoring	8	0	8	0	0	0	158.93
FBD	**Frontend and Backend Development**	**128**	**0**	**128**	**0**	**0**	**0**	**2,542.93**
FBD1	Expense Uploads	16	0	16	0	0	0	317.87
FBD2	Uploaded Expenses Real-time Dashboard	4	0	4	0	0	0	79.47
FBD3	AI-Extracted Expense Data Review and Correction	20	0	20	0	0	0	397.33
FBD4	Uploaded Expenses Deletion	4	0	4	0	0	0	79.47
FBD5	Expense Reports View, Creation, Edit, Submission, and Deletion	20	0	20	0	0	0	397.33
FBD6	Company Email User Authentication	8	0	8	0	0	0	158.93
FBD7	Role-based User Access	16	0	16	0	0	0	317.87

(Continued on next page)

(Continued from previous page)

			Hours per role					
ID	**Name**	**Total hours**	PM	D	T	SA	TW	**Cost (€)**
FBD8	Submitted Report Approval and Rejection	8	0	8	0	0	0	158.93
FBD9	Employee Spending Limit Configuration	8	0	8	0	0	0	158.93
FBD10	Submitted and Reviewed Expenses Record Visualization	4	0	4	0	0	0	79.47
FBD11	Email Notification Service Integration	20	0	20	0	0	0	397.33
BD	**Backend Development**	**120**	**0**	**120**	**0**	**0**	**0**	**2,384.00**
BD1	RESTful API Backend Logic Implementation	20	0	20	0	0	0	397.33
BD2	Azure SQL Database Integration and Manipulation	20	0	20	0	0	0	397.33
BD3	Azure Blob Storage Integration	8	0	8	0	0	0	158.93
BD4	Microsoft Document Intelligence AI Model Integration	20	0	20	0	0	0	397.33
BD5	Automated AI-driven Data Extraction Pipeline	20	0	20	0	0	0	397.33
BD6	Azure SignalR Real-time Notification Service Integration	16	0	16	0	0	0	317.87
BD7	Microsoft Entra ID User Authentication Integration	16	0	16	0	0	0	317.87
FD	**Frontend Development**	**10**	**0**	**10**	**0**	**0**	**0**	**198.67**
FD1	Expenses Search and Filter	8	0	8	0	0	0	158.93
FD2	Reports Search	2	0	2	0	0	0	39.73
Total		**513**	**139**	**401**	**5**	**16**	**41**	**13,370.03**

Table 6.1.2: Personnel cost breakdown by task, role, and total personnel cost. [Own creation]

6.1.2 Generic Cost

In addition to personnel costs, other project costs must also be considered. Therefore, the missing costs are detailed in this section, followed by the formula applied to obtain the resulting cost.

Amortization

Physical resources are also required and used in the project. However, considering only the product purchase price would not be entirely correct. For this reason, the depreciation of each hardware resource has been calculated.

Hardware resources have a 4-year depreciation period. Other relevant factors for amortization calculation include the total project duration and the number of hours dedicated to the project each day, respectively 122 days and 4 hours per day. Since the hardware resources are used throughout the project timeline, their total usage time corresponds to the total project hours (513 hours). Accordingly, the following amortization equation is applied:

$$\text{Amortization (€)} = \text{Resource price} \times \frac{1}{4\text{ years}} \times \frac{1}{122\text{ days}} \times \frac{1}{4\text{ hours}} \times \text{hours used}$$

The amortization cost of each hardware resource is presented in Table 6.1.3.

Hardware Resources	**Price (€)**	**Quantity**	**Total Price (€)**	**Usage Time (h)**	**Amortization (€)**
PC Dell 16 Pro	1,193.53	1	1,193.53	513	313.67
Wireless Mouse	11.99	1	16.99	513	4.47
Dell Monitor	171.82	2	343.64	513	90.31
Keyboard	19.99	1	19.99	513	5.25
Total					**408.44**

Table 6.1.3: Amortization cost of the project's hardware resources. [Own creation]

Electric cost

The current cost of electricity is around **0.1748 €/kWh**. Electricity costs are calculated based solely on the electricity-consuming hardware resources and their respective usage times. Following the same approach used for amortization cost calculation, the usage time is **513 hours**. Table 6.1.4 details the calculations and the resulting total cost:

Hardware Resources	**Power (W)**	**Usage Time (h)**	**Energy Consumed (kWh)**	**Electricity Cost (€)**
PC Dell 16 Pro [**dell-16-pro**]	65	513	33.345	5.83
Wireless Mouse [**logitech-m171**]	0.05	513	0.02565	0.00
Dell Monitor [**dell-pro-24-plus**]	24	513	12.312	2.15
Keyboard [**logitech-k120**]	0.1	513	0.0513	0.01
Total				**7.99**

Table 6.1.4: Electricity cost of the project's hardware resources. [Own creation]